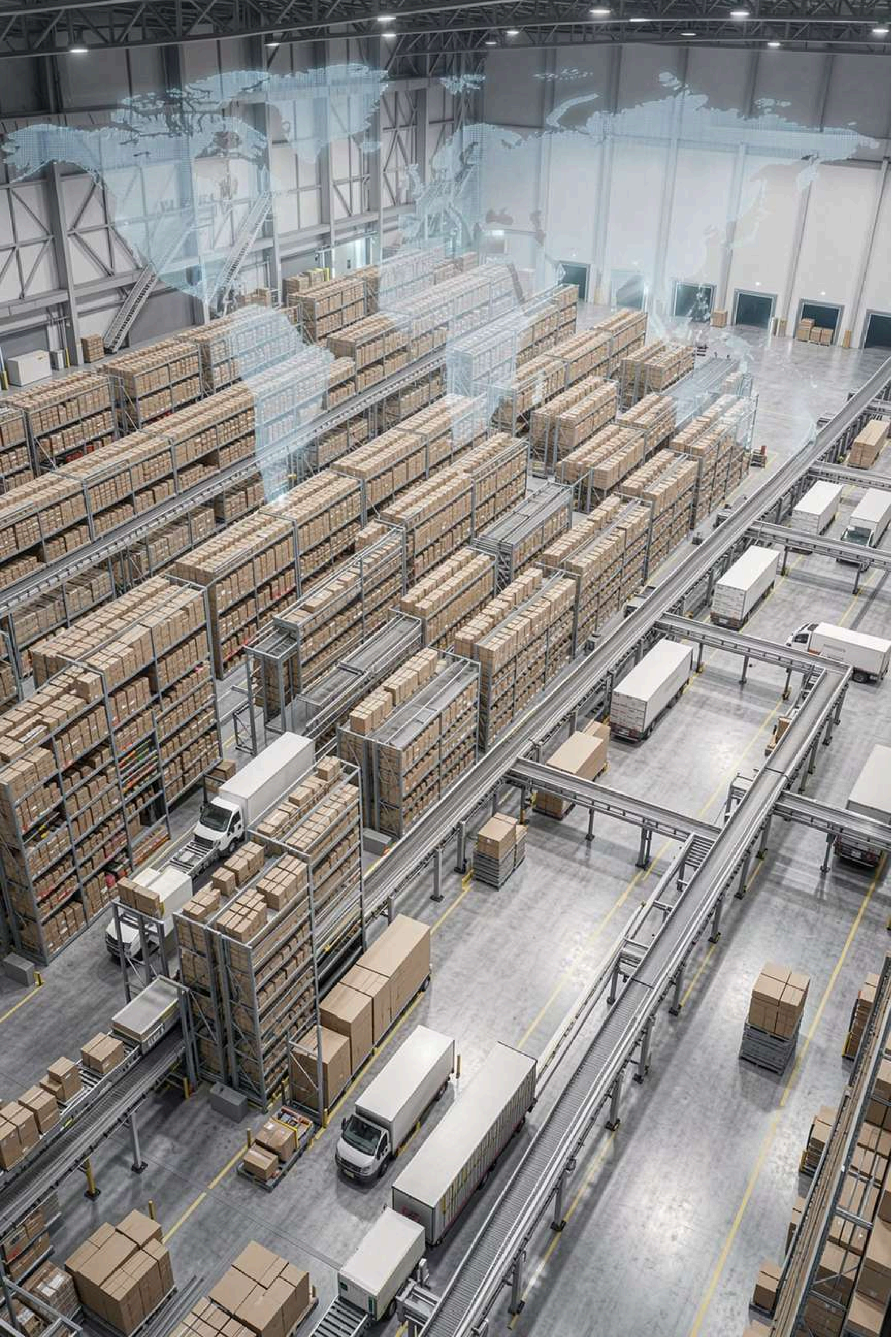


Global Superstore: Sales Performance & Profitability Analysis

CAPSTONE PROJECT 2 — DATA ANALYTICS

PYTHON + TABLEAU

RETAIL & E-COMMERCE

> *How can a global retail company optimize profitability across markets, product categories, and shipping strategies by identifying key revenue drivers, loss-making segments, and regional performance disparities?*

Project Overview

This project performs an **end-to-end data analytics pipeline** on the Global Superstore dataset — **51,290 transactions** across **147 countries** (2011–2014). We clean raw data using Python, conduct exploratory and statistical analysis, build an interactive Tableau dashboard, and deliver actionable business recommendations.

01

Identify

Most and least profitable product categories and sub-categories across global markets

02

Analyze

Shipping cost efficiency and delivery performance by ship mode

03

Evaluate

Regional and country-level sales patterns and profit margins

04

Uncover

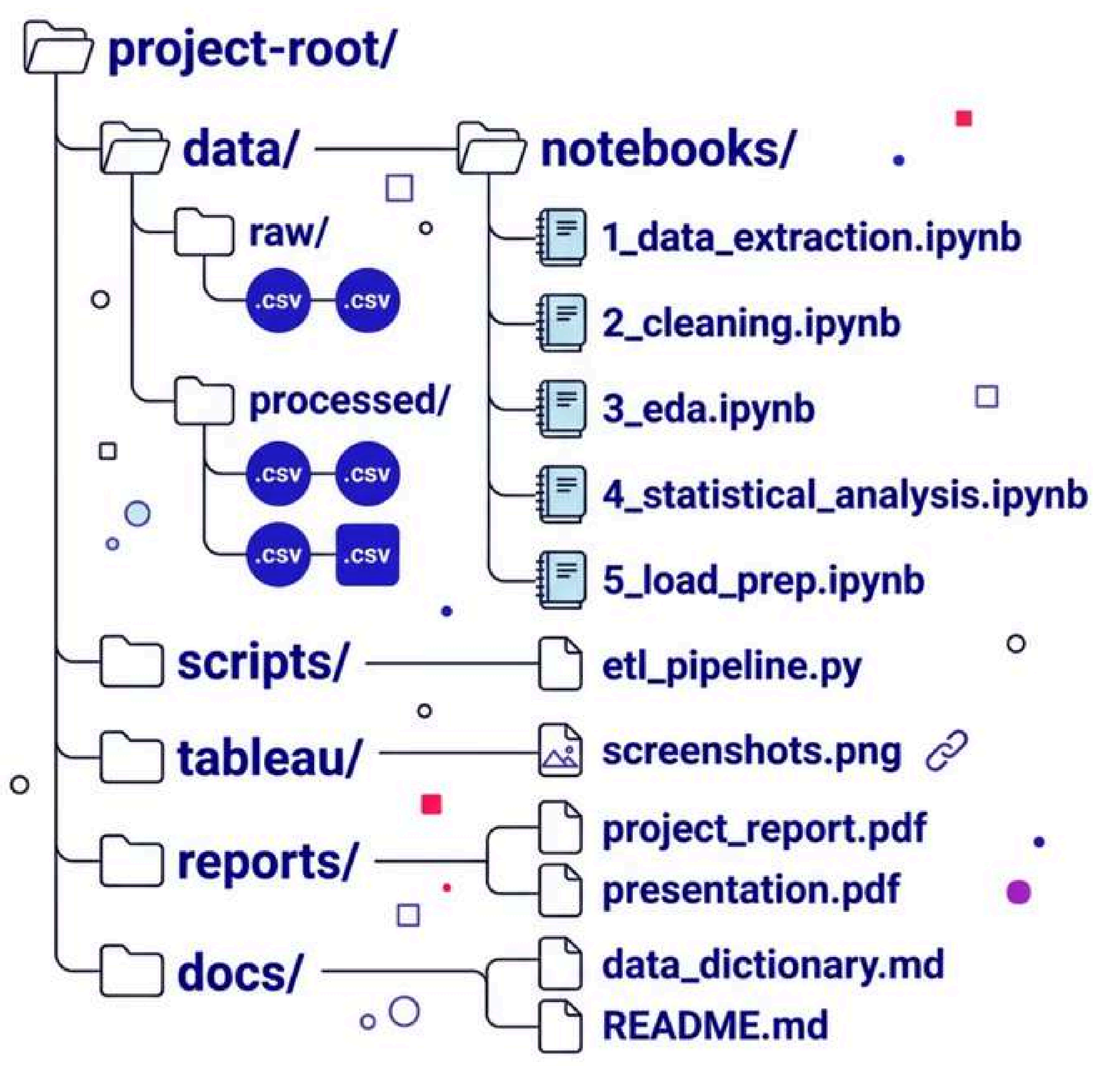
The impact of discounts on profitability

05

Recommend

Data-backed strategies to improve operational efficiency and revenue

Repository Structure



Organized for Reproducibility

The repository follows a clear, sequential structure — from raw data ingestion through cleaning, analysis, and final Tableau-ready export. Each notebook is numbered to reflect the pipeline stage, making the workflow easy to audit and extend.

data/
Raw & cleaned CSV files

notebooks/
5-stage analysis pipeline

scripts/
Standalone ETL script

tableau/
Dashboard & screenshots

reports/
Final report & slides

Dataset at a Glance

51,290

Transactions

Row-level transaction records

27

Columns

Raw dataset attributes

147

Countries

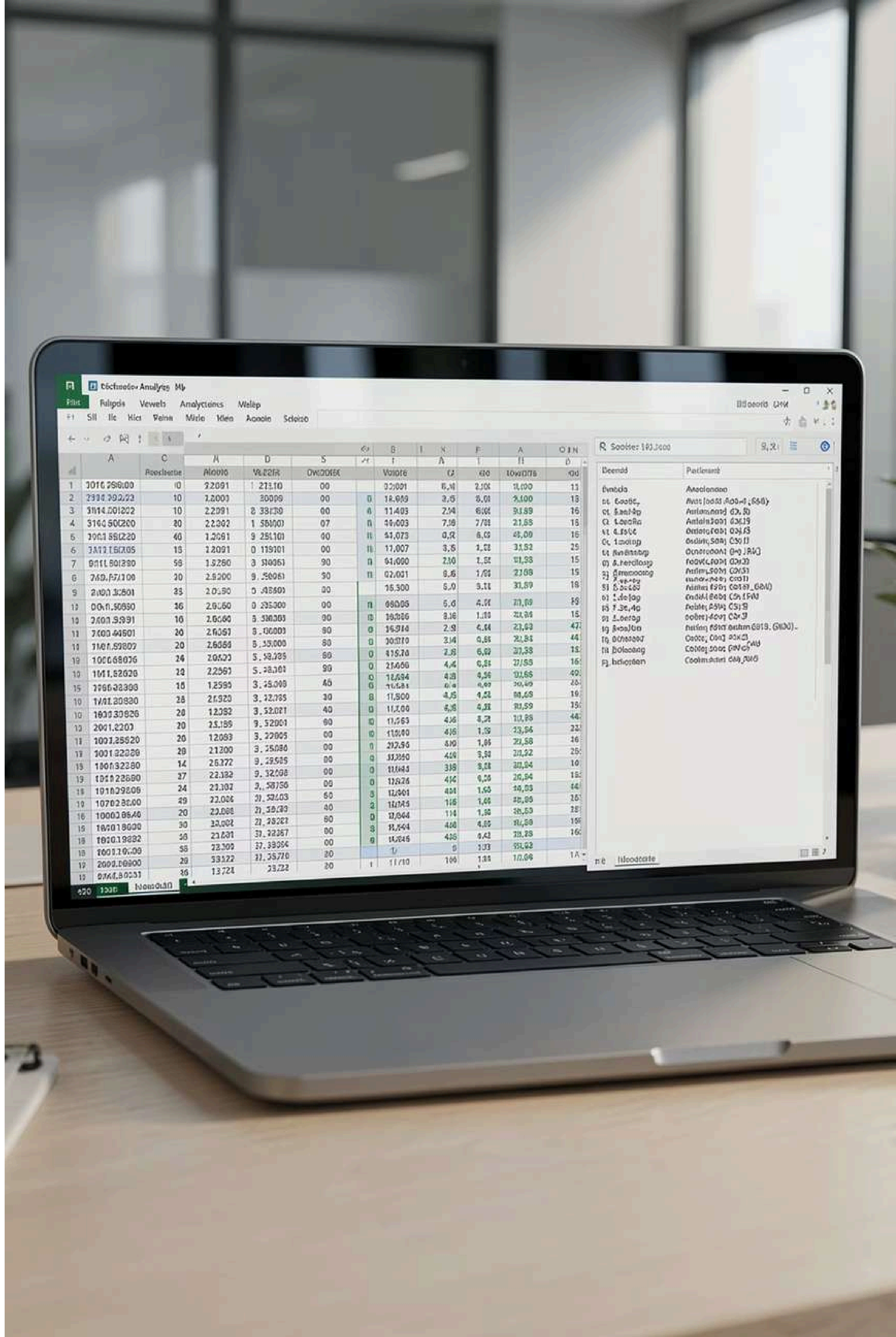
Global market coverage

4

Years

Data span: 2011–2014

⚠ **Quality Issues Found:** Missing values, inconsistent date formats, non-English column headers (记录数), redundant columns, and mixed encoding required comprehensive ETL treatment before analysis.



Tools & Technologies

Python 3.x

Core language for ETL pipeline, data cleaning, and statistical analysis

Pandas / NumPy

Data manipulation, transformation, and numerical computation

Matplotlib / Seaborn

Python-based visualizations for EDA and reporting

SciPy / Statsmodels

ANOVA, chi-square, correlation, and regression testing

Tableau Public

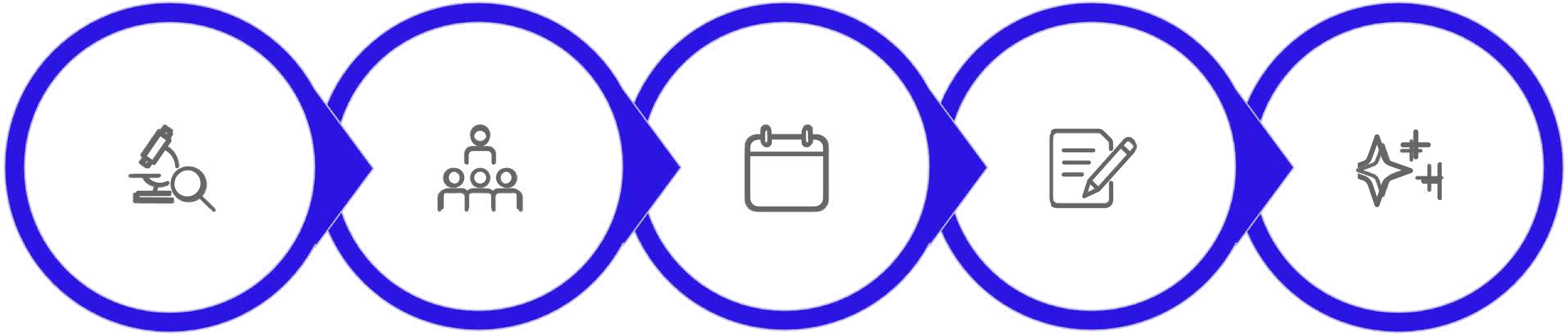
Interactive dashboard design and public publishing

GitHub

Version control, collaboration, and project hosting

ETL Pipeline Summary

A rigorous **7-step ETL pipeline** was applied to transform the raw Kaggle dataset into a clean, analysis-ready format — ensuring data integrity, consistency, and enriched features for downstream modeling.



Profiling | Standardize | Parse Dates | Impute | Feature Eng

Feature engineering added critical derived columns — **Shipping_Days** and **Profit_Margin_%** — enabling deeper profitability and logistics analysis without re-computation.

Key Insights — Preview

Technology Leads Revenue

Highest revenue category, but with volatile profit margins requiring careful monitoring

Tables Are the Biggest Loss-Maker

Consistently unprofitable sub-category across all global markets

Same Day Shipping Is Inefficient

Highest cost-to-sales ratio yet lowest adoption rate globally

Central Africa Surprises

Highest average profit margins despite lower overall sales volumes

Discounts Above 30% Destroy Profit

Strongly correlated with negative profitability across segments

Team Members & Contributions

Harshit Singh

ETL LEAD

Data profiling, cleaning pipeline, GitHub repository setup

Devansh Saini

EDA LEAD

Exploratory data analysis, trend identification, visualizations

Yash Lahase

STATISTICIAN

Statistical tests, regression modeling, hypothesis testing

Sudip Kumar Prasad

VISUALIZATION LEAD

Tableau dashboard design, interactivity, and publishing

Garv Gogna

REPORT LEAD

Final report, business recommendations, and presentation

Project Resources

Explore the Outputs

All project artifacts are organized and accessible within the repository. Use the links below to navigate to each deliverable.

Tableau Dashboard

Interactive visualizations — see tableau/dashboard_links.md

Data Dictionary

Full schema documentation — see docs/data_dictionary.md

Project Report

Detailed findings and methodology — see reports/project_report.pdf

License & Academic Context

This project was developed as part of the **Data Visualization & Analytics Capstone 2** program. It is intended for **academic and educational purposes** only.

Academic Use

Capstone project for a structured data analytics program

Open Dataset

Global Superstore dataset sourced from Kaggle (Rohit Sahoo)

Reproducible

Full pipeline documented in Jupyter notebooks with version control on GitHub